Syngamy

Report:	809a64b7
Generated:	2026-07-31 03:29:19Z
Run started:	not recorded
Software:	OriginMarker 2.1.1 (Kinetochore), Syngamy module
Sperm donor:	GSM4472397_sperm_DNA_71.CEL.txt.gz (825,656 markers, SHA-256 1b56911dae7030f4)
Oocyte donor:	GSM4472417_donor_B_69.CEL.txt.gz (825,656 markers, SHA-256 0b030bbb4bdd5465)
Donor heterozygosity:	16.658% autosomal, of called markers
Samples:	1 analysed (1 gynogenetic)
Assembly:	GRCh37
Data handling:	every file was read in the browser that produced this report. No genotype was transmitted, and none is retained.

Result summary

Sample	Origin	Pat absence	Pat x ceil	Mat absence	Mat x ceil	Zygosity	Sperm	Informative
GSM4774676_mMII-4_55.CEL.txt.gz	gynogenetic	11.11%	9.5x	0.91%	0.8x	uniparental homozygous	-	533,152

Absence is the rate at which the donor's obligate allele is missing where he is homozygous. Ceiling is what this sample's own noise can manufacture: its no-call rate times its heterozygous fraction. A sample is called as lacking a paternal contribution only above 3x that ceiling, and as carrying one only at or below it; in between it is left uncalled. No confidence percentage is reported, deliberately: see Interpretation.

GSM4774676_mMII-4_55.CEL.txt.gz vs GSM4472397_sperm_DNA_71.CEL.txt.gz and GSM4472417_donor_B_69.CEL.txt.gz

GYNOGENETIC

No paternal contribution. Maternal in origin.

paternal absent 11.11% | ceiling 1.17% | 9.5x | 533,152 informative markers | zygosity uniparental homozygous
maternal absent 0.91% | ceiling 1.17% | 0.8x | 537,942 informative markers

The oocyte donor accounts for this genome and the sperm donor does not. Measured against both arrays, so this is not an absence standing in for a presence.

The two declared parents agree at 71.8% of shared markers.

Evidence, sperm donor

Measurement	Observed	Reference	The reference is
Donor alleles absent	11.11%	1.17%	this sample's no-call rate times its heterozygous fraction. Dropout manufactures absence only by turning a heterozygous call homozygous and discarding the donor's allele, so the bound is the product and neither term alone.
Alleles the donor lacks	16.30%	8.33%	half the donor's heterozygosity. A second parent supplies an allele he lacks only where he is homozygous, at sum(pq), and his heterozygosity is sum(2pq).
Heterozygous BAF band	3.95%	8.00%	measured at 15-16% for diploid genomes and 1.3-3.4% for uniparental ones. Dropout does not mimic this: a biparental embryo at 33% dropout still reads 22.2%.

Evidence, oocyte donor

Measurement	Observed	Reference	The reference is
Donor alleles absent	0.91%	1.17%	this sample's no-call rate times its heterozygous fraction, computed the same way as for the sperm donor. The two rates are comparable only because the instrument is the same.
Alleles the donor lacks	8.08%	8.48%	half the oocyte donor's heterozygosity, which is what a second parent contributes.

Sex chromosomes

Chr	Informative	Absent	Rate	x ceiling	Verdict	Note
chrX	16,161	3,157	19.53%	16.6	absent	

Chromosomes, sperm donor

Chr	Informative	Absent	Rate	x ceiling	Verdict	Note
chr1	44,183	4,724	10.69%	9.1	absent	
chr2	42,259	4,545	10.76%	9.2	absent	
chr3	35,801	3,685	10.29%	8.8	absent	
chr4	32,321	3,613	11.18%	9.5	absent	
chr5	30,064	3,428	11.40%	9.7	absent	
chr6	37,270	4,732	12.70%	10.8	absent	
chr7	28,828	2,901	10.06%	8.6	absent	
chr8	26,160	3,004	11.48%	9.8	absent	
chr9	23,358	2,615	11.20%	9.5	absent	
chrX	16,161	3,157	19.53%	16.6	absent	
chr10	26,249	2,984	11.37%	9.7	absent	
chr11	26,293	3,042	11.57%	9.9	absent	
chr12	24,168	2,713	11.23%	9.6	absent	
chr13	17,947	2,143	11.94%	10.2	absent	
chr14	18,402	2,094	11.38%	9.7	absent	
chr15	16,875	2,104	12.47%	10.6	absent	

Research use only. Parent of origin here is a statistical call from array genotypes and requires confirmation by an independent method in a qualified genetics laboratory. Not a clinical diagnostic.

Chr	Informative	Absent	Rate	x ceiling	Verdict	Note
chr16	19,401	2,158	11.12%	9.5	absent	
chr17	20,165	2,005	9.94%	8.5	absent	
chr18	15,075	1,888	12.52%	10.7	absent	
chr19	18,753	1,664	8.87%	7.6	absent	
chr20	13,214	1,426	10.79%	9.2	absent	
chr21	7,730	915	11.84%	10.1	absent	
chr22	8,636	872	10.10%	8.6	absent	

Chromosomes, oocyte donor

Chr	Informative	Absent	Rate	x ceiling	Verdict	Note
chr1	44,435	337	0.76%	0.6	present	
chr2	42,879	266	0.62%	0.5	present	
chr3	36,093	231	0.64%	0.5	present	
chr4	32,297	202	0.63%	0.5	present	
chr5	29,864	249	0.83%	0.7	present	
chr6	37,020	223	0.60%	0.5	present	
chr7	29,525	255	0.86%	0.7	present	
chr8	25,941	184	0.71%	0.6	present	
chr9	23,592	227	0.96%	0.8	present	
chrX	13,463	110	0.82%	0.7	present	
chr10	26,441	167	0.63%	0.5	present	
chr11	26,552	588	2.21%	1.9	unclear	
chr12	24,169	248	1.03%	0.9	present	
chr13	18,030	195	1.08%	0.9	present	
chr14	18,216	133	0.73%	0.6	present	
chr15	16,865	108	0.64%	0.5	present	
chr16	19,867	343	1.73%	1.5	unclear	
chr17	20,918	214	1.02%	0.9	present	
chr18	14,899	83	0.56%	0.5	present	
chr19	19,601	352	1.80%	1.5	unclear	
chr20	13,844	113	0.82%	0.7	present	
chr21	7,918	61	0.77%	0.7	present	
chr22	8,976	91	1.01%	0.9	present	

Sample quality

Markers:	825,656 read, 684,832 called (82.9%)
No-call:	17.06%
Heterozygous:	8.93% of called
chrX / autosomal het:	1.024
Sex call:	female
Product:	UK Biobank Axiom Array
Assembly:	GRCh37 (0 placements illegal under GRCh37, 6,315 under GRCh38, 825,656 tested)
Allele coding:	ok BAF corroborates 0=AA, 2=BB
Mean BAF AA/AB/BB:	0.044 / 0.533 / 0.966
SHA-256:	09790d5e6a060b511a57191470409ce96067d13ad53ba72e68f195150c6d75a3
File size:	63,468,993 bytes

Quality gates

Gate	Value	Verdict	Basis
call rate	82.9%	usable	inside the 75-95% amplified usable band; the Axiom vendor gate of 97% would reject this sample, and that gate was established on unamplified bulk DNA
het-to-hom asymmetry valid	82.9%	usable	holds within the usable call-rate band
genome-wide LOH	8.9%	report only	no genome-wide LOH signature
heterozygosity plausible	8.9%	usable	within the plausible range for one diploid genome
numeric genotype coding	-	usable	BAF corroborates 0=AA, 2=BB
sex call	102.4%	usable	chrX/autosome heterozygosity ratio 1.024 is inside the female band. Dropout cancels in the ratio, so this survives amplified material.

Findings

No contribution from this parent anywhere. chrY alone would not have separated this from an X-bearing sperm carrying a full paternal genome.
Every allele traces to this parent and the genome is homozygous: a parent-only complement, duplicated. Nothing here required chrY.
No chrY, yet the parent's alleles are present on chrX as well as the autosomes: an X-bearing sperm. This is the case a chrY test cannot call.

Research use only. Parent of origin here is a statistical call from array genotypes and requires confirmation by an independent method in a qualified genetics laboratory. Not a clinical diagnostic.

Methods

Parent of origin was determined from SNP array genotypes using OriginMarker 2.1.1 (Kinetochore). For each of 1 sample(s), the rate at which the sperm donor's obligate allele was absent was computed across autosomal markers where he is homozygous and the sample is called, and compared against the rate that sample's own genotyping noise can produce, taken as the product of its no-call rate and its heterozygous fraction. A sample was called as lacking a paternal contribution only where the observed rate exceeded that bound by 3-fold, and as carrying one only where it fell at or below the bound; rates between the two were left uncalled. Whether a second parent contributed was assessed from the rate at which the sample carries alleles the sperm donor does not possess, against half his heterozygosity, which is the contribution a second parent makes. Zygosity was taken from the fraction of B-allele frequencies falling in the heterozygous band, 0.35 to 0.65. Sperm type was inferred from the chrX rate rather than from chrY, so that an X-bearing sperm carrying a full paternal genome is distinguishable from an absent paternal contribution; neither case leaves a Y.

An oocyte donor array was supplied, so each sample was tested against both parents by the same measurement and the class read off the pair. This separates a sample lacking a paternal contribution from one belonging to neither declared parent, which a single parent cannot do: with the sperm donor alone a maternal origin is the shape left behind by his absence rather than a presence that was measured. The two parents were also compared against each other, since two arrays of one person pass both tests and yield a confident biparental call.

Assembly was determined from marker positions against the UCSC chromInfo and gap tables and found to be GRCh37. No liftOver was performed.

Interpretation

No confidence percentage is attached to a call, and this is deliberate. The per-sample likelihood ratio between "paternal genome present" and "absent" runs past 10^10000 on a genome-wide array, which is not a probability any reader should be handed. The honest figure is the empirical accuracy of the method, and on the pairs of known relationship it has been run against that is 9 of 9 correct, whose 95% lower bound is roughly 72%. About 300 consecutive correct calls would be needed to claim 99%. What is reported instead is the margin: the observed rate, the reference it was measured against, and the ratio between them, so a reader can see how far from the boundary the call sits.

The absence axis separates by roughly thirty-fold and carries the call. The alleles-the-donor-lacks axis separates by about 1.6-fold and only distinguishes a biparental genome from a paternal-only one that is heterozygous; treat a near-boundary call on that axis as provisional. A sample scoring in the uncalled band is not a weak positive, it is an absence of evidence: an array from the second parent measures dropout directly and settles it.

Constants and where they come from

Constant	Value	Provenance
Mendelian inconsistency floor	0.0063	Kothiyal 2019, 0.63% of variant sites across 1,314 nuclear families. A lower bound on the spurious-violation rate, never the value itself.
Paternal-absence noise bound	no-call rate x heterozygous fraction	Derived per sample, not fitted. Bounds every related pair measured, within 2x: 0.13% predicted against 0.05% observed, 0.18% against 0.16%, 10.4% against 9.69%.
Absence calling margin	3x	How far above the noise bound an absence must sit before it is called.
Residual absence floor	0.005	Absence on clean data from genotyping error alone, measured at 0.03% and 0.05%. Added to the bound so a near-perfect array still has a non-zero reference.
Second-parent signal	half the donor's heterozygosity	Derived. A second parent supplies an allele the donor lacks only where he is homozygous, at sum(pq); his heterozygosity is sum(2pq) over the same markers.
Diploid heterozygous BAF band	0.08 of markers in 0.35-0.65	Measured at 15-16% for diploid genomes and 1.3-3.4% for uniparental ones. Dropout does not mimic this: a biparental embryo at 33% dropout still reads 22.2%.
Per-chromosome minimum	200 informative markers	Below this a chromosome is not reported: the rate is too noisy to place against the ceiling.
Assembly tables	UCSC chromInfo and gap, hg19 and hg38	Freely redistributable. No liftOver chain file is used or shipped: those carry a non-commercial field-of-use restriction Apache 2.0 cannot sublicense.

Files read

File	Role	Bytes	Markers	SHA-256
GSM4472397_sperm_DNA_71.CEL.txt.gz	donor	63,674,062	825,656	1b56911dae7030f4ecebc2ade9c7b1bacee23def81d130e6b3d0ac9edb388169
GSM4472417_donor_B_69.CEL.txt.gz	oocyte	62,633,502	825,656	0b030bbb4bdd5465e01f9d4b67d9ca13940435ae0c768ca6889ff7345adf0ae1
GSM4774676_mMII-4_55.CEL.txt.gz	sample	63,468,993	825,656	09790d5e6a060b511a57191470409ce96067d13ad53ba72e68f195150c6d75a3

The hash is of the file as read, so a reviewer can confirm the input without being sent it.

Citation

OriginMarker 2.1.1 (Kinetochore). Syngamy: parent of origin from SNP array genotypes. Report 809a64b7, generated 2026-07-31 03:29:19Z.
Kothiyal P et al. Mendelian inconsistent signatures from 1,314 ancestrally diverse family trios. J Comput Biol 2019;26(5):405-419. doi:10.1089/cmb.2018.0253. Cited for the inconsistency floor above.